15 Three solid shapes **A**, **B** and **C** are similar.

The surface area of shape **A** is $4\,cm^2$
The surface area of shape **B** is $25\,cm^2$

The ratio of the volume of shape **B** to the volume of shape **C** is $27:64$

Work out the ratio of the height of shape **A** to the height of shape **C**.
Give your answer in its simplest form.

.....

(Total for Question 15 is 4 marks)

16 Prove algebraically that $0.2\dot{5}\dot{6}$ can be written as $\frac{127}{495}$

Let $x = 0.2565656...$

$1000x = 256.565656...$

1 mark
Show the repeating block clearly

$10x = 2.565656...$

Subtract: $990x = 254$

1 mark
Matching decimal tails cancel

$x = \frac{254}{990} = \frac{127}{495}$

1 mark
Divide numerator and denominator by 2

Therefore the recurring decimal equals $\frac{127}{495}$

(Total for Question 16 is 3 marks)

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